

Q Which are the major organisations that have become members/investors in this initiative? Are there others in talks for coming on board?

The investors in the fund are corporates, banks, insurers, and funds of funds - including SIDBI through its Fund of Funds for Startups (FFS) programme, ICICI Lombard, Philips, Bajaj Electricals Ltd and RBL Bank, among others.

Ray Stata, founder and chairman of Analog Devices, who has been an icon of innovation for the last several decades, has also joined the Fund as an advisor and investor.

Q What milestones has this venture reached so far and what are your immediate plans?

We have publicly announced investments in three startups so far. The Bharat Innovation Fund did its first formal close in July 2018 and immediately started investing. We have publicly announced three investments - in Entropik, Detect Technologies and CreditVidya.

We are in the final stages of doing the due diligence for a few more startups which will be announced shortly. We feel encouraged by the sheer quality of deals and the type of IP-led innovation we have come across in healthcare, fintech, industrial and in overall digital transformation.

Q Tell us about the startups funded till now.

Of the three start-ups we've invested in so far, let's look at Entropik Technologies. It is an emotion AI company that provides insights on a consumer's subconscious responses to external stimuli. The company's proprietary solution, Affect Lab, is an online emotional intelligence platform that combines EEG brainwave mapping, facial coding and eye tracking technologies that can be used by brands and content creators for end-to-end consumer research, with long-term potential in creating an emotion AI database.

Detect Technologies is a startup that originated in the IIT Madras incubator. It uses a proprietary magnetostrictor material sensor and advanced image processing algorithms to determine defects in high temperature oil pipelines. It also provides software and algorithms built on top of third party drones, for the inspection of hard-to-reach assets in difficult climatic conditions and in inaccessible terrain.

CreditVidya is a startup that is strengthening the lending market in India by automating the underwriting process in digital lending through a machine learning-enabled credit risk assessment platform. It uses alternative data in addition to bureau information. The technology is implemented through software integration into the lenders' workflow process, assisting in loan disbursal.

Q How deeply does BIF collaborate with those in Indian academia?

BIF is built on the decade-long startup engagement experience of CIIE, IIM Ahmedabad, which has seeded over 200 startups across its various programmes such as iAccelerator, Power of Ideas, Powerstart, India Innovation Growth Program, and Startup Oasis, to name a few.

We work closely with many academic startup incubators at several IITs, at the IISc and other institutions. We also are in touch with top academicians from universities and research institutions. We collaborate with them to build our investment thesis, and routinely reach out to technical experts to understand the new technologies and IPs of startups that seek us for their funding needs.

Q Which verticals, according to you, provide the best business opportunities for electronics and tech startups today?

For startups, a purely hardware play might prove difficult in a super competitive landscape. But a hardware-software combination that captures data from the physical world and then uses AI/ML to build analytics solutions is where the opportunity for Indian electronics and

technology startups lies.

While there are opportunities to innovate in the electronics sector alone (with sensors, bio-printing, AI/ML hardware accelerators, etc), the pool of such innovative startups from India is relatively small and they need to compete on these fundamental technologies globally. But the opportunity we see is in the hardware-software combined play that leverages the software strengths of the Indian ecosystem.

Q What do you think are the biggest challenges product-based technology startups face today?

The biggest challenges faced by product-based hardware driven startups in India are the lack of facilities for prototyping and design for manufacturability, as well as the relative dearth of an overall electronics manufacturing ecosystem. Iterations take time and access to quality vendors in India is still a challenge.

The government initiatives of value addition, although improving the ecosystem, still have a long way to go. Many have ventured out to China to access this ecosystem but relatively smaller quantities and not being physically present has hindered their efforts. The challenge is also lack of funding for these startups in the subsequent rounds and access to early homegrown customers who are willing to adopt the technology or solution.

Q What are the solutions to these challenges?

The India Electronics and Semiconductor Association (IESA) and IoTForum, a think tank working on building the IoT ecosystem, are focused on potential solutions for these challenges. For example, IESA has now started a fabless initiative with the government of Karnataka to help incubate the fabless semiconductor startups by giving them access to EDA tools and funding test chips. There are also several hardware-only incubators and accelerators like Forge, NASSCOM IoT and Revvx that are focused on bridging this gap. Large